THE NUMBER OF BRANCHES OF AN ALGEBRAIC SPACE AT A POINT COUNTED ON ETALE LOCAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original lemma 04KF: sources/stacks-chow/tags.txt, source lines 5971–5971. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0CB4: sources/stacks-chow/tags.txt, source lines 15459–15459. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $x \in |X|$ be a point. Let $n \in \{1, 2, \ldots\}$ be an integer. The following are equivalent*

1. *for any scheme U and étale morphism $a : U \to X$ and $u \in U$ with $a(u) = x$ the number of minimal primes of the local ring $\mathcal{O}_{U,u}$ is $\leq n$ and for at least one choice of U, a, u it is n,*
2. *for any scheme U and étale morphism $a : U \to X$ and $u \in U$ with $a(u) = x$ the number irreducible components of U passing through u is $\leq n$ and for at least one choice of U, a, u it is n,*
3. *for any scheme U and étale morphism $a : U \to X$ and $u \in U$ with $a(u) = x$ the number of branches of U at u is $\leq n$ and for at least one choice of U, a, u it is n,*
4. *for any scheme U and étale morphism $a : U \to X$ and $u \in U$ with $a(u) = x$ the number of geometric branches of U at u is n, and*
5. *the number of minimal prime ideals of $\mathcal{O}_{X,\overline{x}}$ is n.*

Proof. The equivalence of (1) and (2) follows from the fact that irreducible components of U passing through u are in 1-1 correspondence with minimal primes of the local ring of U at u. Let $a : U \to X$ and $u \in U$ be as in (1). Then $\mathcal{O}_{X,\overline{x}}$ is the strict henselization of $\mathcal{O}_{U,u}$ by Lemma 04KF. Recall that the (geometric) number of branches of U at u is the number of minimal prime ideals of the (strict)

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

henselization of $\mathcal{O}_{U,u}$. In particular (4) and (5) are equivalent. The equivalence of (2), (3), and (4) follows from More on Morphisms, Lemma 0CB4. $\square$